

HF

Ch 9. The Gaussian channel, $X_i \rightarrow \oplus \rightarrow Y_i$

$$Y_i = X_i + Z_i$$

$\uparrow$ $\uparrow$ $\uparrow$
 output input noise

Z_i is i.i.d from Gaussian distribution with variance N .

• That is, $Y_i = X_i + Z_i$, $Z_i \sim N(0, N)$. Further assume Z_i is indep of X_i .

Here, without further condition, the capacity of this channel may be ∞ .

• Thus, we assume an average power constraints:

For any codeword $(X_1, \dots, X_n)$ transmitted over the channel we require that

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \leq P$$

• the additive noise in the above channel may be due to a variety of cause.

• By central theorems the Gaussian is reasonable in a large number scheme.

9.1 Gaussian channel: Definitions.

Def The information capacity of the Gaussian channel with P is

$$C = \max_{\text{all } E[X^2] \leq P} I(X, Y)$$

(since Z is independent of X)

$\downarrow$

$$\text{then, } I(X, Y) = h(Y) - h(Y|X) = h(Y) - h(Z|X) \stackrel{(\text{since } Z \text{ is independent of } X)}{=} h(Y) - h(Z)$$

$$\text{Now, } h(Z) = \frac{1}{2} \cdot \log 2\pi e N$$

$$\text{Also, } E[Y^2] = E[(X+Z)^2] = E[X^2] + 2E[X \cdot Z] + E[Z^2] = P + N$$

$$\therefore \text{ (given } E[Y^2] = P + N, \text{ the entropy of } Y \leq \frac{1}{2} \cdot \log 2\pi e (P + N)$$

by the fact that the normal maximizes the entropy for a given variance.

∴ By using $(*)$ and the above, we have

$$I(X; Y) = h(Y) - h(Z) \leq \frac{1}{2} \log 2\pi e(P+N) - \frac{1}{2} \log 2\pi eN = \frac{1}{2} \log \left(1 + \frac{P}{N}\right)$$

$\Rightarrow C = \max_{X^2 \leq P} I(X; Y) = \frac{1}{2} \log \left(1 + \frac{P}{N}\right)$, where the maximum is attained when $X \sim N(0, P)$.

Def An (M, n) code for the Gaussian channel with P consist of the following

1. An index set $\{1, \dots, M\}$
2. An encoding fn $x: [1 \dots M] \rightarrow X^n$ yielding codewords $x^{(n)}(1), \dots, x^{(n)}(M)$ satisfying the P , that is

$$\forall \text{ codeword } \sum_{i=1}^n x_i^2(w) \leq nP, \quad w = 1, \dots, M.$$
3. A decoding fn $g: Y^n \rightarrow \{1, \dots, M\}$.

Note the arithmetic average of the probability error is defined by

$$P_e^{(n)} = \frac{1}{2^{nR}} \sum_{i=1}^{2^{nR}} p_i$$

Def A rate R is achievable for a Gaussian channel with P if there exists a sequence $(2^{nR}, n)$ codes with codewords satisfying the power constraint s.t the maximal probability of error $P_e^{(n)} \rightarrow 0$.

9.2 The supremum of the achievable rates for the Gaussian channel.

• $R \leq C$ (Goel)

pf) We must show that if $P_e^{(n)} \rightarrow 0$ for a sequence $(2^{nR}, n)$ codes for a Gaussian channel with $P \Rightarrow R \leq C = \frac{1}{2} \cdot \log \left(1 + \frac{P}{N}\right)$.

Consider any $(2^{nR}, n)$ code that satisfies the P.C,

$\frac{1}{n} \sum_{i=1}^n x_i^2(w) \leq P$ ($1 \leq w \leq 2^{nR}$) and let W be distributed uniformly over $\{1, \dots, 2^{nR}\}$. Then,

the uniformly distribution $W \in \{1, \dots, 2^{nR}\} \Rightarrow$ a distribution on the input code words $\Rightarrow$ a distribution over the input alphabet.

$$W \rightarrow \hat{x}^{(n)}(w) \rightarrow Y^n \rightarrow \hat{w}.$$

Use Fano's inequality to get $H(W) \leq 1 + n R \cdot P_e^{(n)} = n \Sigma_n$, where $\Sigma_n \rightarrow 0$ as $P_e^{(n)} \rightarrow 0$.

$$\therefore nR = H(W) = I(W; \hat{W}) + H(W | \hat{W}) \leq I(W; \hat{W}) + n \Sigma_n = h(Y^n) - h(Y^n | X_1^n) + n \Sigma_n$$

$$\stackrel{(*)}{\leq} \sum_{i=1}^n h(Y_i) - h(Z^n) + n \Sigma_n = \sum_{i=1}^n h(Y_i) - \sum_{i=1}^n h(Z_i) + n \Sigma_n$$

$$\stackrel{(*)'}{=} \sum_{i=1}^n I(X_i; Y_i) + n \Sigma_n$$

Next, let P_i be the average power of the i -th column of the code

$$\left(P_i = \frac{1}{2nR} \sum_{\omega} X_i^2(\omega) \right). \text{ Note that } \mathbb{E} Y_i^2 = P_i + N.$$

Since entropy is maximized by the normal distribution, we have that

$$h(Y_i) \leq \frac{1}{2} \cdot \log 2\pi e (P_i + N) \quad \text{--- (7)}$$

$$\text{By combining (7) with } (*)' \Rightarrow nR \leq \sum (h(Y_i) - h(Z_i)) + n \Sigma_n$$

$$\leq \sum \left(\frac{1}{2} \log (2\pi e (P_i + N)) - \frac{1}{2} \log 2\pi e N \right) + n \Sigma_n = \sum \frac{1}{2} \log \left(1 + \frac{P_i}{N} \right) + n \Sigma_n$$

From $\frac{1}{n} \sum_i P_i \leq P$ and $f(x) = \frac{1}{2} \log(1+x)$ is concave, it follows that

$$\frac{1}{n} \cdot \sum_{i=1}^n \frac{1}{2} \log \left(1 + \frac{P_i}{N} \right) \leq \frac{1}{2} \cdot \log \left(1 + \frac{1}{n} \sum_{i=1}^n \frac{P_i}{N} \right) \leq \frac{1}{2} \log \left(1 + \frac{P}{N} \right).$$

$$\therefore R \leq \frac{1}{2} \cdot \log \left(1 + \frac{P}{N} \right) + \Sigma_n. \quad \therefore \text{we have the desired result by } \Sigma_n \rightarrow 0$$

9.3 Bandlimited channel

Def Bandlimited channel with white noise is a continuous time channel

$$\text{Let } \underset{\substack{\uparrow \\ \text{output}}}{y(t)} = (\underset{\substack{\uparrow \\ \text{signal wave form}}}{x(t)} + \underset{\substack{\uparrow \\ \text{noise term of white Gaussian channel}}}{z(t)}) \quad \rightarrow \text{impulse response}$$

Goal: How to calculate the capacity of such a channel?

To be precise, how to show that sampling a bandlimited signal at a sampling rate $\frac{1}{2W}$ is sufficient to reconstruct the signal from the samples.

Then suppose that a t.c.n $f(t)$ is bandlimited to W , namely, the spectrum (Fourier transform) of the t.c.n is 0 for all frequencies ω greater than W
 $\Rightarrow$ the t.c.n is completely determined by samples of the t.c.n spaced $\frac{1}{2W}$ seconds apart.

pf) The plot in book is wrong.

Let $F(\omega)$ be the Fourier transform of $f(t)$. $\Rightarrow f(t) = \frac{1}{2\pi} \int_{-2\pi W}^{2\pi W} F\left(\frac{\omega}{2\pi}\right) \cdot e^{i\omega t} d\omega$
 ($\because F\left(\frac{\omega}{2\pi}\right)$ is 0 outside the band $-2\pi W \leq \omega \leq 2\pi W$)
 $\therefore f\left(\frac{n}{2W}\right) = \frac{1}{2\pi} \int_{-2\pi W}^{2\pi W} F\left(\frac{\omega}{2\pi}\right) \cdot e^{i\omega n} d\omega \quad (n \in \mathbb{Z})$

The sample values $f\left(\frac{n}{2W}\right)$ determine the Fourier coefficients and by extension, they determine the value of $F(\omega)$ in the interval $(-W, W)$.

Since a function is uniquely determined by its Fourier transform and since $F(\omega)$ is 0 outside the band W , we can determine the function from the samples (uniquely). Now, consider

⊛ $\text{sinc}(t) = \frac{\sin(2\pi W t)}{2\pi W t} \Rightarrow$ the spectrum of this function is constant in $(-W, W)$ and is zero outside this band.

(Refer to Rudin's book. ch 10)

Now, define $g(t) = \sum_{n=-\infty}^{\infty} f\left(\frac{n}{2W}\right) \text{sinc}\left(t - \frac{n}{2W}\right)$.

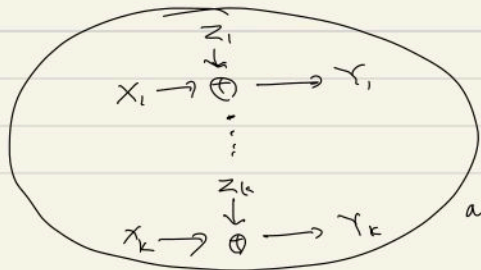
$\Rightarrow$ It follows that $g(t)$ is bandlimited to W and is equal to $f\left(\frac{n}{2W}\right)$.

at $t = \frac{n}{2W}$. Thus, by the ⊛ we get $f(t) = g(t)$.

9.4 parallel Gaussian channel.

Consider K independent Gaussian channels in parallel with a common P .

(Goal): how to distribute the total power among the channels so as to maximize the capacity with nonwhite Gaussian noise where each parallel component represent a different frequency.



First, assume that we have a set of Gaussian channels:

$Y_j = X_j + Z_j, \quad j = 1, \dots, K$ with $Z_j \sim N(0, N_j)$

and noise is independent from channel to channel.

given $E \sum_{j=1}^K X_j^2 \leq P$.

• the information capacity of the channel $C = \max_{\substack{p(x_1, \dots, x_k) \\ \sum E x_i^2 \leq P}} I(x_1, \dots, x_k; y_1, \dots, y_k)$

• Since $z_1, \dots, z_k$ are independent, $I(x_1, \dots, x_k; y_1, \dots, y_k)$

$$\begin{aligned} &= h(y_1, \dots, y_k) - h(y_1, \dots, y_k | x_1, \dots, x_k) = h(y_1, \dots, y_k) - h(z_1, \dots, z_k | x_1, \dots, x_k) \\ &= h(y_1, \dots, y_k) - h(z_1, \dots, z_k) = h(y_1, \dots, y_k) - \sum_i h(z_i) \leq \\ &\leq \sum_i (h(y_i) - h(z_i)) \leq \sum_i \frac{1}{2} \cdot \log \left(1 + \frac{p_i}{N_i} \right), \text{ where } p_i = E x_i^2 \\ &\text{and } \sum p_i = P. \quad (\text{Equality holds when } (x_1, \dots, x_k) \sim \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} p_1 & 0 \\ 0 & p_k \end{bmatrix} \right)) \end{aligned}$$

↓ our problem is converted to

maximize $\frac{1}{2} \cdot \sum_i \cdot \log \left(1 + \frac{p_i}{N_i} \right)$ given $\sum p_i = P$.

Use KKT condition to obtain $p_i = (V - N_i)^+$ under $\sum (V - N_i)^+ = P$.

9.5 channels with colored Gaussian noise.

Goal: we will consider the case when the noise is dependent. To be precise, we can consider a block n -consecutive uses of the channel as n channels in parallel with dependent noise.

• Let K_z be the covariance matrix of the noise
 K_x be the " " " input.

• the power constraint : $\frac{1}{n} \cdot \sum E x_i^2 \leq P \iff \frac{1}{n} \cdot \text{tr}(K_x) \leq P$.

• $I(x_1, \dots, x_n; y_1, \dots, y_n) = h(y_1, \dots, y_n) - \underbrace{h(z_1, \dots, z_n)}$

→ $[x_i]$ determined only by their distribution, not distribution of input

∴ We hope to maximize $h(y_1, \dots, y_n)$

For this, recall the entropy is maximized when y is normal, which is achieved when the input is normal. Also, $K_y = K_x + K_z$ and the entropy $h(y_1, \dots, y_n) = \frac{1}{2} \cdot \log((2\pi e)^n |K_x + K_z|)$, $| \cdot |$ is det.

↓

choosing K_X given $\frac{1}{n} \cdot \text{tr}(K_X) \leq P$, $\therefore$ our problem is reduced to

$K_Z = Q \Lambda Q^T$ $\rightarrow$ diagonal matrix

since $Q \cdot Q^T = I \Rightarrow |K_X + K_Z| = |K_X + Q \Lambda Q^T| = |Q| |Q^T K_X Q + \Lambda| |Q^T|$
 $= |Q^T K_X Q + \Lambda| = |A + \Lambda|$, where $A = Q^T K_X Q$, and then
 $\text{tr}(A) = \text{tr}(Q^T K_X Q) = \text{tr}(Q^T Q K_X) = \text{Tr}(K_X)$.

$\therefore$ the above problem is reduced to

maximizing $|A + \Lambda|$ subject to $\text{tr}(A) \leq nP$.

By Hadamard's inequality, we have that

$|A + \Lambda| \leq \prod_i K_{ii}$, with equality holds $\Leftrightarrow A + \Lambda$ is diagonal.

$\therefore |A + \Lambda| \leq \prod_i (A_{ii} + \lambda_i)$, $\Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$, with equality

holds $\Leftrightarrow A$ is diagonal. Power constraint $\Leftrightarrow \frac{1}{n} \cdot \sum_i A_{ii} \leq P$, $A_{ii} \geq 0$

Use KKT condition to yield $A_{ii} = (v - \lambda_i)^+$, where v is chosen so that $\sum A_{ii} = nP$.

$\therefore$ Thus, this value A maximizes the entropy of Y and hence, the mutual information.

9.6 Gaussian channel with feedback (FB)

• We consider channels with memory when the noise is correlated from time instant to time instant. Feedback does increase capacity.

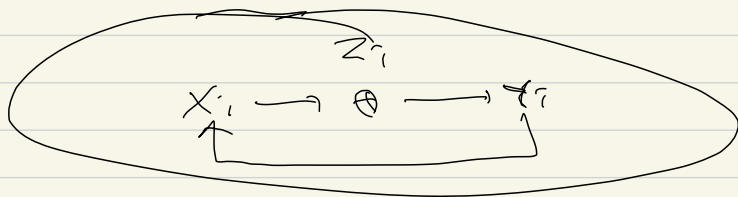
• We describe an expression for the capacity in terms of the covariance matrix of the noise Z .

Goal: ① To prove a converse for this expression for capacity

② simple bound on the increase in capacity due to FB.

• the Gaussian channel with FB is $y_i = x_i + z_i$, $z_i \sim N(0, K_2^{(n)})$

Note A $(2^{nR}, n)$ for the Gaussian channel with FB consist of a sequence of mappings $x_i(W, y^{i-1})$, where $w \in \{1, \dots, 2^{nR}\}$ is the input message and y^{i-1} is a sequence of past values of the output.



• A power constraint: $E \left[\frac{1}{n} \sum_{i=1}^n x_i^2(W, y^{i-1}) \right] \leq P$, ($1 \leq W \leq 2^{nR}$)
where the expectation is over all possible noise sequence

Caution. X^n and Z^n are not independent and X_i depends on the past values of Z .

• FB, without FB.

① with FB: The capacity $C_{n,FB}$ bits per transmission of the time-varying Gaussian channel with FB is

$$C_{n,FB} = \max_{\frac{1}{n} \text{tr}(K_X^{(n)}) \leq P} \frac{1}{2n} \log \frac{|K_{X+Z}^{(n)}|}{|K_Z^{(n)}|} \quad (*)$$

where the maximization is taken over all X^n of the form

$$x_i = \sum_{j=1}^{i-1} \underbrace{(b_{ij})}_{(FB)} z_j + v_i \quad (1 \leq i \leq n)$$

(b_{ij}) (innovation process) and V^n is indep of Z^n .

② with FB, The capacity C_n of the time-varying Gaussian channel with feedback IS

$$C_n = \max_{\substack{K_x^{(n)} \\ \frac{1}{n} \text{tr}(K_x^{(n)}) \leq 1}} \frac{1}{2n} \log \frac{|K_x^{(n)} + K_z^{(n)}|}{|K_z^{(n)}|}$$

Then For a Gaussian channel with FB, the rate R_n for any sequence of $(2^{nR_n}, n)$ codes with $P_e^{(n)} \rightarrow 0$ satisfies $R_n \leq C_{n,FB} + \epsilon_n$ with $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$, where $C_{n,FB}$ is defined as (A)'

pt) Let w be uniform over 2^{nR} and use Fano's Ineq to have that $H(w|\hat{w}) \leq 1 + nR_n \cdot P_e^{(n)} = n\epsilon_n$ (where $\epsilon_n \rightarrow 0$ as $P_e^{(n)} \rightarrow 0$)

$$\Rightarrow nR_n = H(w) = I(w; \hat{w}) + H(w|\hat{w}) \leq I(w; \hat{w}) + n\epsilon_n$$

$$\stackrel{\uparrow}{\leq} I(w; Y^n) + n\epsilon_n = \sum (w; Y_i | Y^{i-1}) + n\epsilon_n$$

(by data processing Ineq in ch 2)

$$\begin{aligned} &= \sum (H(Y_i | Y^{i-1})) - H(Y_i | W, Y^{i-1}, X_i, X^{i-1}, Z^{i-1}) + n\epsilon_n \\ &= \sum (H(Y_i | Y^{i-1})) - H(Z_i | Z^{i-1}) + n\epsilon_n \quad \text{(} H(Y^n) - H(Z^n) + n\epsilon_n \text{)} \end{aligned}$$

then, divide n to yield $R_n \leq \frac{1}{n} (H(Y^n) - H(Z^n)) + \epsilon_n$.

$$\Rightarrow \leq \frac{1}{2n} \log \frac{|K_Y^{(n)}|}{|K_Z^{(n)}|} + \epsilon_n \leq C_{n,FB} + \epsilon_n. \quad \square$$

Q how to estimate the bdd on the capacity with FB and without FB in terms of $K_x^{(n)}, K_z^{(n)}$??

Lemma 1 Let X and Z be n -dimensional random vectors

$$\Rightarrow K_{X+Z} + K_{X-Z} = 2K_X + 2K_Z.$$

Lemma 2 For $n \times n$, $A, B \succeq 0$, if $A - B \succeq 0$

$$\Rightarrow |A| \succeq |B|.$$

Lemma 3 For $n \times n$ $x, z \succeq 0 \Rightarrow |K_{x+z}| \leq 2^n \cdot |K_x + K_z|$

Lemma 4 For $n \times n$ $A, B \succeq 0$, and $0 \leq \lambda \leq 1$

$$\Rightarrow |\lambda A + (1-\lambda)B| \succeq |A|^\lambda \cdot |B|^{1-\lambda}.$$

Def x^n is causally related to z^n if
 $f(x^n, z^n) = f(z^n) \cdot \prod_{i=1}^n f(x_i | x^{i-1}, z^{i-1})$.

Lemma 5 If x^n and z^n are causally related $\Rightarrow$
 $h(x^n - z^n) \geq h(z^n)$.

Then $C_{n, FB} \leq C_n + \frac{1}{2}$.

pt) Combining all the results, we have

$$C_{n, FB} = \max_{x, K_x \in \mathcal{N}} \frac{1}{2^n} \log \frac{|K_x|}{|K_z|} \leq \max_{x, K_x \in \mathcal{N}} \frac{2^n |K_x + K_z|}{|K_z|}$$

$$= \max_{x, K_x \in \mathcal{N}} \frac{1}{2^n} \cdot \log \frac{|K_x + K_z|}{|K_z|} + \frac{1}{2} \stackrel{\text{(Lemma 3)}}{\leq} C_n + \frac{1}{2}. \quad \square$$

$\hookrightarrow$ (def of capacity with FB).

Then $C_{n, FB} \leq 2 \cdot C_n$. $\left(\Leftrightarrow \frac{1}{2} \cdot C_{n, FB} \leq C_n \right)$

pt) WTS $\frac{1}{2} \frac{1}{2^n} \log \frac{|K_x + K_z|}{|K_z|} \leq \frac{1}{2^n} \log \frac{|K_x + K_z|}{|K_z|}$

Note that $\frac{1}{2^n} \log \frac{|K_x + K_z|}{|K_z|} = \frac{1}{2^n} \log \left(\frac{1}{2} |K_x + K_z| + \frac{1}{2} |K_x + K_z| \right)$

(First Lemma)

$$\geq \frac{1}{2^n} \cdot \log \frac{|K_x|^{\frac{1}{2}} |K_z|^{\frac{1}{2}}}{|K_z|} \stackrel{\text{(Lemma 4)}}{\geq} \frac{1}{2^n} \log \frac{|K_x|^{\frac{1}{2}} \cdot |K_z|^{\frac{1}{2}}}{|K_z|} = \frac{1}{2} \cdot \frac{1}{2^n} \cdot \log \frac{|K_x|}{|K_z|}$$

(Lemma 4)

(Lemma 5)

we sample this part



Thank for.